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Development tendency analysis for the water resource carrying capacity based on system dynamics model and the improved fuzzy comprehensive evaluation method in the Changchun city, China

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ABSTRACT

Water resource carrying capacity describes the relationship between different systems such as the socio-economic, water environment, and water resources systems. Recent studies tend to focus on the qualitative description of water resource carrying capacity, with a lack of quantitative research on the topic. This study proposes an improved water resource carrying capacity assessment method based on combining the improved fuzzy comprehensive evaluation and the system dynamics model, enabling the quantitative and qualitative measurement of water resource carrying capacity. This method can dynamically compute the water resource carrying capacity under different social development scenarios through the feedback regulation and system simulation of social economy–water resources and the water environment interactions. A comprehensive assessment of trends from six scenarios between 2018 and 2025 was conducted, based on scenarios designed for different purposes. The results indicate that if the current development model is adopted, the water resource carrying capacity will continue to decline and remain at a low level of "normal carrying" (0.431 in 2025). However, the water resource carrying capacity in Changchun may improve by changing the production mode of the national economy and adjusting the proportion of the economy. Rationally allocating water resources and strengthening water ecological protections can significantly increase the water resource carrying capacity and maintain it at "positive carrying", while maintaining steady economic and social growth.

1. Introduction

Since the beginning of the 21st century, rapid economic development and exponential population growth has led to the consumption of large quantities of water resources. With an increase in the demand for water resources, water shortages have become an obstacle to the sustainable development of the social economy (Lin et al., 2005). And water pollution problems have also deteriorated the health of ecosystems, impacting on the development of the social economy (Zhu et al., 2010). Therefore, the development of hydropower and other renewable energy sources with little impact on the environment is of great significance to achieving sustainable development (Kuriqi et al., 2019a). For example, the implementation of hydrologically based environmental flow methods can help alleviate water shortages and ecological environmental problems in the downstream reaches (Kuriqi et al., 2019b). Research into the water resource carrying capacity (WRCC) has been gaining attention in efforts to address such predicaments and achieve sustainable development.

The water resource carrying capacity is the maximum affordability of economic development when coordinated development of the economy and society is achieved under the principle of sustainable development. Although the WRCC concept was proposed relatively late, it has developed rapidly. In the early stages, the influence of water resource abundance on the WRCC was often singularly considered. For example, Joardar (1998) considered the impact of water supply on the WRCC, and

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Nomenclature

Abbreviations and acronyms

WRCC	Water resource carrying capacity
IFCE	Improved fuzzy comprehensive evaluation
SD	System dynamics
FCE	Fuzzy comprehensive evaluation
AHP	Analytical hierarchical process
GDP	Gross domestic product
COD	Chemical oxygen demand
FAHF	Forest, animal husbandry and fishery

Harris and Kennedy (1999) used the water resources agricultural carrying capacity as an indicator to measure regional development potential. Recently, it has been recognized that the issue of water shortage and security has a growing impact on economic development. The research on the WRCC has transformed from the examination of a simple natural factor to the study of the water–ecological–economic nexus (Ren et al., 2016; Liu et al., 2017; Yue et al., 2015; Ait-Aoudia and Berezowska-Azzag, 2016).

There are currently many research methods used to study the WRCC, mainly focusing on index evaluation methods and system dynamics methods (Wang et al., 2017; Zhou et al., 2017). The index evaluation method has become a generally accepted and widely used method due to its simple operation. For example, the traditional trend analysis method (Song et al., 2011), the principal component analysis method (Cao et al., 2020), the multi-objective analysis method (Wang et al., 2016), the artificial neural network method (Luo et al., 2019), and the fuzzy comprehensive evaluation (FCE) method. As the traditional trend analysis method lacks systematic analysis of the interaction between various factors, the research results only reflect the general trends in the WRCC, where it is difficult to complete an objective description of the actual situation. Zhang et al (2019) applied the principal component analysis method to evaluate the temporal scale variation tendency of the WRCC. However, there was still some uncertainty when integrating the method WRCC index standardization, the method of principal component determination, and the weights of contribution rates. The multi-objective analysis is influenced by its limitations of scale and is more suitable for smaller areas of research. The artificial neural network prediction is based on relatively independent systems, which means it is challenging to consider the coupling between systems. The FCE method is a method widely used by scholars (Zhang et al., 2020). While the FCE method can solve problems associated with qualitative evaluation, it is constrained by the variable weights when multiple factors are considered and restricted by directional evaluation (Gong and Jin, 2009). In the beginning, most scholars habitually used the expert evaluation method to determine the index weight in the index system. This makes the selection of indicator weights more subjective (Wang et al., 2020; Miao et al., 2019). Therefore, scholars have successively used the analytical hierarchical process (AHP), entropy method and other weight determination methods to make up for the deficiencies of the FCE method (Chi et al., 2019; Wu and Hu, 2020). On the basis of a comprehensive study of the literature, a method that combines the AHP and FCE method (called an improved fuzzy comprehensive evaluation method) is selected to obtain an objective qualitative evaluation. In contrast, the indicator evaluation methods focus on static trend analysis, while the SD method achieves system simulation via the dynamic feedback between various influencing factors of each subsystem (Zhang et al., 2014). The SD method may be used to investigate the behaviors of a complex system over time by converting the whole system into an interconnected series of stocks and flows, affecting each other through feedback loops (Zomorodian et al., 2018). The biggest difference between SD method and other methods is its own negative feedback system, that is, a readjustment of constraint conditions. This kind of constraint can constantly change over time, so as to achieve a real dynamic simulation. Because it is good at solving practical problems with nonlinearity, uncertainty and dynamics, the SD method is considered widely (Ali et al., 2019; Yang et al., 2015; Yang et al., 2019). However, the SD method also has its limitations, such as with regard to qualitative evaluation. The SD method can integrate numerical and descriptive techniques, systematically investigate and evaluate complex systems, and realize different internal relations for water resources systems (Mashaly and Fernald, 2020). A few scholars tried to combine the principal component analysis method or AHP method with the SD model and obtained good simulation results (Dai et al., 2020; Yang et al., 2019). Compared with simple models or index evaluations, these coupled methods can solve the problem of qualitatively and quantitatively evaluating the regional WRCC.

Thus, this study proposes a method for evaluating the WRCC that combines the improved fuzzy comprehensive evaluation (IFCE) method with the SD model. Fig. 1 shows a flowchart of the methodology proposed in this study. The integration of this method enables the qualitative and quantitative analysis to provide a basis for fully understanding regional development trends. This study establishes a coupling relationship model consisting of five subsystems: economic, population, water resources demand, water supply, and water environment subsystems. Through the addition of restrictive factors, the true conditions of the WRCC's system in Changchun City are restored as much as possible. Six reasonable scenarios were proposed to simulate the WRCC development trend in the study area under different scenarios.

2. Study area

Changchun (124°18′–127°05′E, 43°05′–45°15′N) is the capital of Jilin Province and the most well-known old industrial city in northeast China, spanning an area of 2.06×10^4 km^2 (Fig. 2). Changchun's mean annual precipitation is 587 mm, and the mean annual total water resource is 3.13 billion m^3. By 2017, the population of Changchun was 7.49 million and the water resource per capita was only 490 m^3, less than a quarter of the national average. This demonstrates that Changchun has been a city experiencing serious water shortages. The rapid development of the industrial and agricultural economy in recent years has triggered a number of associated changes in the social and ecological environment, having specific impacts on the water resources system. According to statistical reports, approximately 4.96×10^6 t wastewater is discharged annually into water bodies, whilst the city achieves a sewage treatment rate of only 83%. The water shortage in terms of volume and quality restricts the sustainable development of economy and society in Changchun. At present, Changchun is the leading development area of the Changji economic circle and an important gateway to northeast Asia. As such, a top priority in the region is understanding how to ensure the coordinated development of the social economy, water resources, and water environment.

3. Data sources and methods

3.1. Statistical data

This research used data on the socio-economics, water resource supply and demand, and water environment. The main data sources include the Jilin statistical yearbook (2005–2017), the Changchun statistical yearbook (2005–2017), and the Changchun water resource bulletin (2005–2017). The water quota reference for the industry water quota standards of Jilin province and some of the environmental data was also sourced from the Changchun Environmental Protection Bureau (2005–2017).

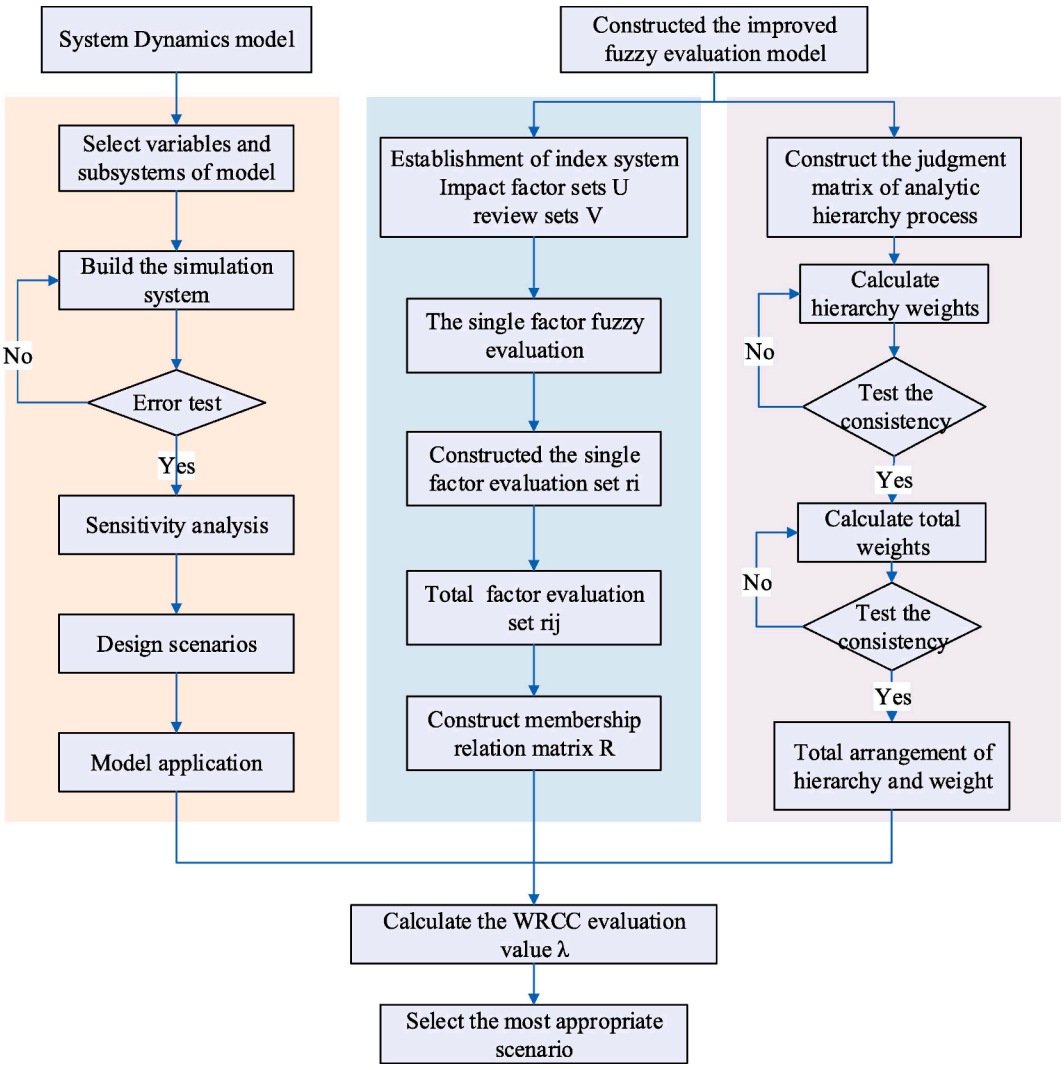


Fig. 1. Flowchart of the methodology.

3.2. The improved fuzzy comprehensive evaluation method

A number of representative evaluation indicators must be selected to obtain a high-quality WRCC evaluation system. As the WRCC evaluation index standard and the boundary of the evaluation system are usually uncertain and fuzzy, the FCE method can increase the objectivity and accuracy of the evaluation results. However, the traditional FCE method often considers the expert's experience as a value in weight selection, which is thus greatly affected by human factors; the analytical hierarchical process (AHP) can just make up for these shortcomings. This study combines AHP and FCE (herein referred as IFCE) to evaluate Changchun's WRCC in efforts to generate more realistic evaluation results. The specific method was as follows:

(1) Establish impact factor sets $U = (u_1, u_2, \ldots, u_m)$ and review sets $V = (v_1, v_2, \ldots, v_n)$, separately. Obtain review set values based on the local government's development plan and current development level (the selection of review sets V, is shown in Table 2).

(2) Use the single factor fuzzy evaluation to evaluate the influencing factors u_i and obtain the single factor evaluation set $r_i = (r_{i1}, r_{i2}, r_{i3})$. Then, determine the membership of each influencing factor individually to generate the matrix, R.

$$R = \begin{bmatrix} r_{11} & r_{12} & \cdots & r_{1n} \\ r_{21} & r_{22} & \cdots & r_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ r_{m1} & r_{m2} & \cdots & r_{mn} \end{bmatrix} \quad (1)$$

(3) Determine the weight of each evaluation indicator using AHP. First, clarify the relationship between the indicators and establish a corresponding indicator system. The indicator system is divided into three levels: the target, criteria, and basic indicator layers. The judgment matrix B is obtained according to the subordinate relationships. The final evaluation weight A is obtained based on the judgement criteria (Table 1):

$$B = (b_{ij})_{m \times m} \quad (2)$$

$$A = (a_1, a_2, \ldots, a_m) = b_j \times c_{ji} \quad (3)$$

where m is number of evaluation factors, and b_{ij} is the ratio of the factor U_i to the importance of U_j relative to a criterion. In addition, $i \neq j$; $i, j = 1, 2, 3, \ldots, n$.

Conduct a consistency check to ensure the reliability of the weight calculation results. As the study area is located in China's famous old industrial base, several industry-related indicators were selected in the evaluation index selection: these were U_{11} and U_{16} (the selected evaluation indexes and their consistency check results are in Table 2). The

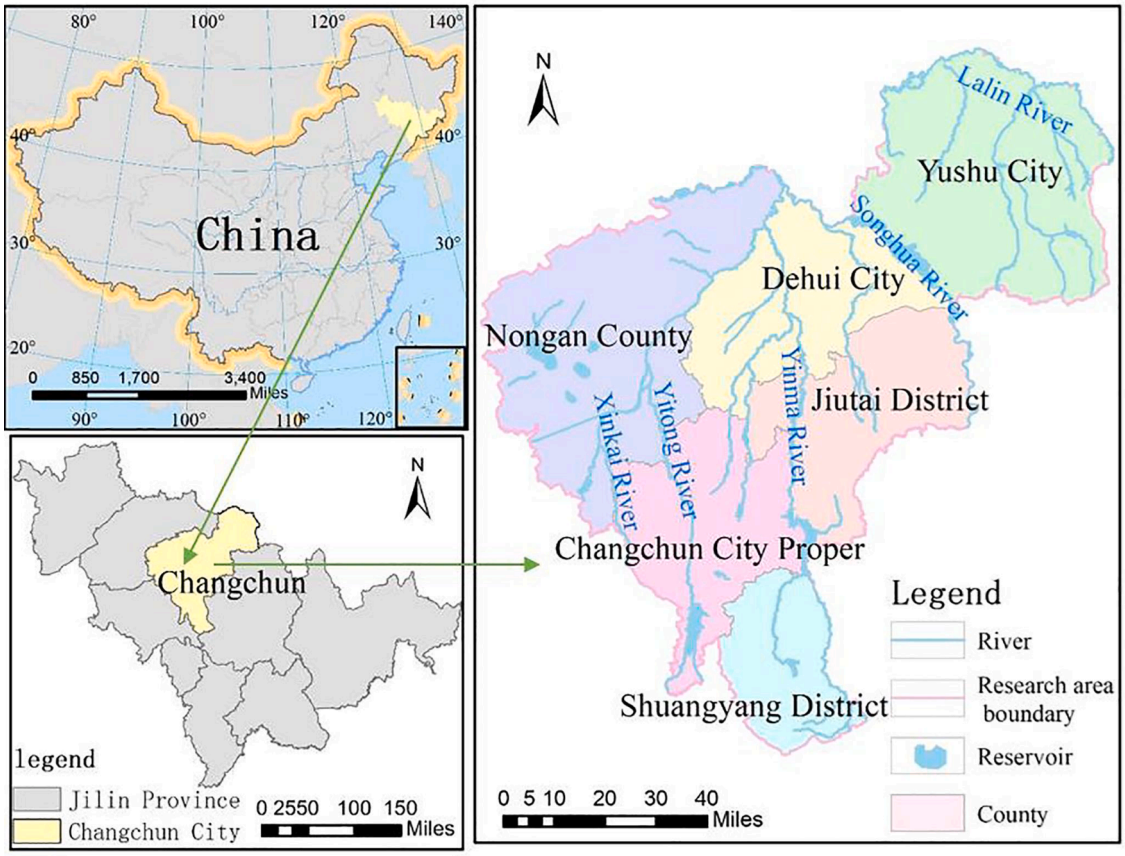


Fig. 2. Location of Changchun and its major rivers.

Table 1
Scale of preference between two elements.

Numerical scale	Definition
1	Equal significance between the two factor
3	Slight significance of one factor compared to the other
5	Strong significance of one factor compared to the other
7	Dominance of one factor over the other
9	Absolute dominance of one factor over the other
2, 4, 6, 8	Intermediate values between two neighboring levels
Reciprocals	If the ratio of the importance of factors i and j is a_{ij}, then the ratio of the importance of j to i is a_{ij}.

consistency check is expressed in Eqs. (4) and (5):

$$CI = \frac{\lambda \max - n}{n-1} \tag{4}$$

$$CR = \frac{CI}{RI} \tag{5}$$

where CI is the consistency index; λ_{max} is the maximum eigenvalue of the judgment matrix; and CR is the consistency ratio. The consistent performance of the judgment matrix was only accepted when CR < 0.1.

(4) The vectorization evaluation result C is obtained from the membership matrix and the weight. To facilitate comparative evaluation, each membership degree C_i, is normalized to obtain a comprehensive score λ:

$$C = (c_1, c_2, \ldots, c_m) = A \times R \tag{6}$$

$$\lambda = \frac{\sum_{i=1}^{n=3} c_i^k \cdot \lambda_i}{\sum_{i=1}^{n=3} c_i^k} \tag{7}$$

3.3. The system dynamics method

SD is a computer-aided approach based on an analytical information and feedback system. SD understands a problem based on the close relationship between the system behavior and the internal mechanism, obtained through the established mathematical model and dynamic feedback process. The distinct advantage of SD is that it can handle high-order, non-linear, multi-feedback and complex time variant system problems (Sterman, 1994). The variable SD type includes a level variable (L), a rate variable (R), and an auxiliary variable (A), where variables are quantitatively described by differential equations (Reference Eqs. (8) and (9)). The corresponding flow chart is established through a large number of variables and their complex feedback relationships, and simulation is performed subsequently:

$$L(t) = L_0 + \int_0^t \left(\sum R_1(t) - \sum R_0(t)\right) dt \tag{8}$$

$$R(t_i) = \frac{L(t_{i+1}) - L(t_i)}{DT} \tag{9}$$

where L(t) in Eq. (8) is the value of level variable L at time, t; L_0 is the initial value of L; R_1 and R_0 are the input and output flow of L, respectively; $R(t_i)$ in Eq. (9) is the rate variable at time, t_i; $L(t_{i+1})$ and $L(t_i)$ are the values of the level variable at time, t_{i+1} and t_i, respectively; and DT is the time interval between time, t_{i+1} and t_i.

Table 2
Selection of evaluation indicators and their important parameters.

Target layer	System level	Evaluation indicator	Indicator description	Weight	CR	Review sets V		
						V_1	V_2	V_3
WRCC	Economic subsystem	U_1 The gross domestic product (GDP) per capita	Reflects the level of people's economic development	0.065	0.0226	>6	4 ~ 6	<4
		U_2 The proportion of primary industry economy	Reflects the level of primary industry economic development	0.039		<6	6 ~ 15	>15
		U_3 The proportion of secondary industry economy	Reflects the level of secondary industry economic development	0.065		<49	49–55	>55
		U_4 The proportion of tertiary industry economy	Reflects the level of tertiary industry economic development	0.027		>45	40 ~ 45	<40
	Social subsystem	U_5 urbanization rate	Reflects the level of urbanization	0.029	0.0265	>50	35 ~ 50	<35
		U_6 available water resource per capita	Reflects the amount of available waterresources	0.123		>1200	400 ~ 1200	<400
		U_7 water supply quantity per capita	Reflects the amount of water supply capacity	0.079		>1000	500 ~ 1000	<500
		U_8 Modulus of water demand	Reflects the amount of regional water demand capacity	0.045		<10	10 ~ 40	>40
	Water resources subsystem	U_9 the rate of water recirculation	Reflects the water resources reuse capacity	0.022	0.0294	>50	35 ~ 50	<35
		U_{10} rate of water supply to water demand	Reflects the relationship of water supply to water demand	0.143		>1.05	1.05 ~ 0.95	<0.95
		U_{11} water consumption per 10,000 RMB of industrial production	Reflects the water consumption of regional industry	0.055		<15	15 ~ 25	>25
		U_{12} water consumption per 10,000 RMB of GDP	Reflects the regional water consumption	0.045		<55	55 ~ 90	>90
		U_{13} water resource utilization rate	Reflects the effective development and utilization of water resources	0.126		<40	40 ~ 90	>90
	Water environment subsystem	U_{14} ratio of water pollution to water demand	Reflects the impact of water pollution on water supply and demand	0.059	0.0246	<0.05	0.05 ~ 0.1	>0.1
		U_{15}Sewage treatment rate	Reflects the capacity of sewage treatment	0.049		>60	40 ~ 60	<40
		U_{16} COD emission per 10,000 RMB of industrial production	Reflects the pollution status of the industry's water environment	0.018		<50	50 ~ 200	>200
		U_{17}Ammonia emission per capita(t/10^4person)	Reflects the pollution status per capita of the water environment	0.012		<5	5 ~ 10	>10

4. Establishing a coupled model based on SD and the improved FCE method

4.1. Establishment of the WRCC model

The WRCC is the result of the joint control of multiple factors. Traditional analysis methods largely focus on the analysis and prediction of a single indicator, thereby only roughly predicting the WRCC for a specific area. These methods do not consider the interaction and dynamic adjustment between multiple factors, and as such, they cannot meet the requirements of simulation systems. This study builds an innovative coupling model based on the SD and IFCE method. It considers the interrelationships and dynamic feedback between various systems, where research results are more aligned with reality. The model designed five closely related subsystems: the economic, population, water resources demand, water supply, and water environment subsystems. The study selected nine state variables, 12 rate variables, 63 auxiliary variables, and 38 table functions, with a total of 122 variables used to build the simulation system. The flowcharts of each subsystem and the representative formula operations between variables are shown in Figs. 3–7 and Table 3. Each subsystem is described in Sections 4.1.1–4.1.5.

4.1.1. Economic subsystem

Economic development is the driving force underpinning the entire water resources system, directly affecting the rate of regional

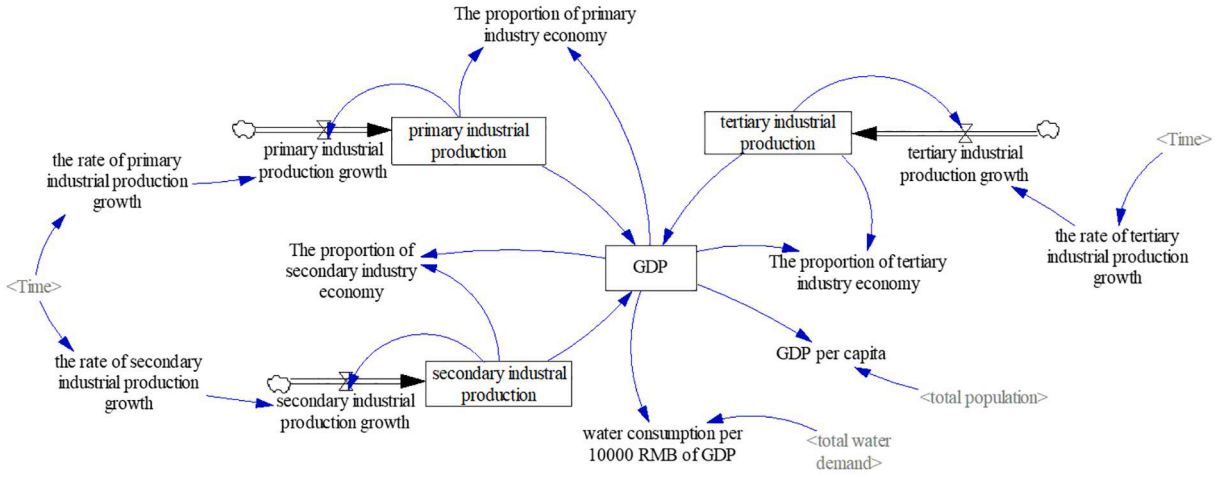


Fig. 3. System dynamics flowchart of the economic subsystem.

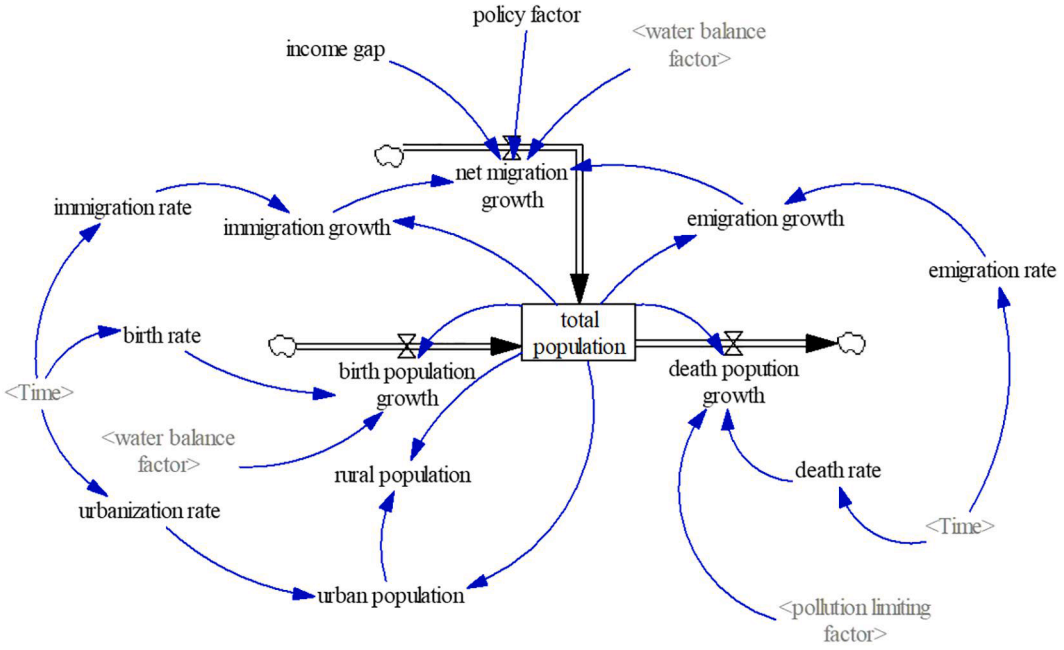


Fig. 4. System dynamics flowchart of the population subsystem.

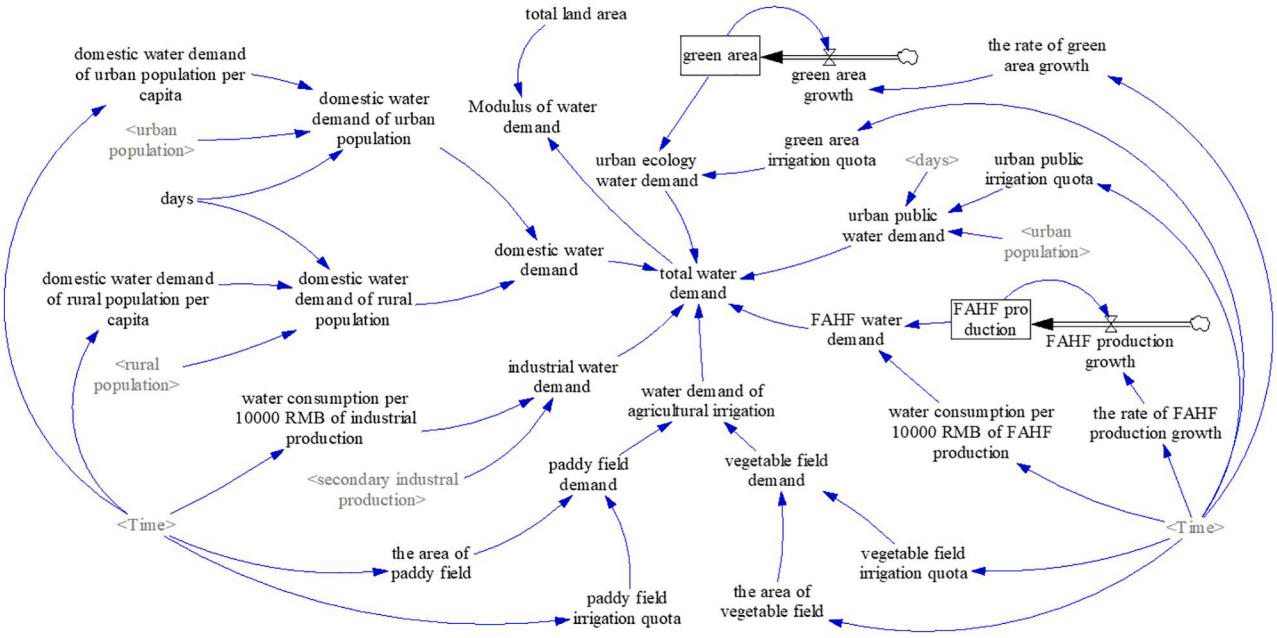


Fig. 5. System dynamics flowchart of water resources demand subsystem.

development and thus the balance of water supply, water demand, and the state of the water environment. In today's globalized world, there is a greater need to ensure rapid economic development, while maintaining the health development of the water resources and the water environment.

4.1.2. Population subsystem

Population changes directly affect changes in the supply and demand of water, whilst in turn changes in available water resources also influence the population. This forms the corresponding feedback adjustment. The labor force also plays a role in the regional economy and ecological environment.

4.1.3. Water resources demand subsystem

The key to determining an accurate WRCC is to understand the relationship between regional water demand and water supply, as unbalanced supply and demand can seriously weaken the regional WRCC. This study analyzes Changchun's water demand across six sectors: the industrial, agricultural irrigation and domestic water, forest, animal husbandry and fishery (FAHF), urban public and urban ecological water demands. By establishing multi-level feedback correlations, the most realistic water demand subsystem is simulated to achieve model simulation effects.

4.1.4. Water supply subsystem

The water supply subsystem determines the supply capacity of water

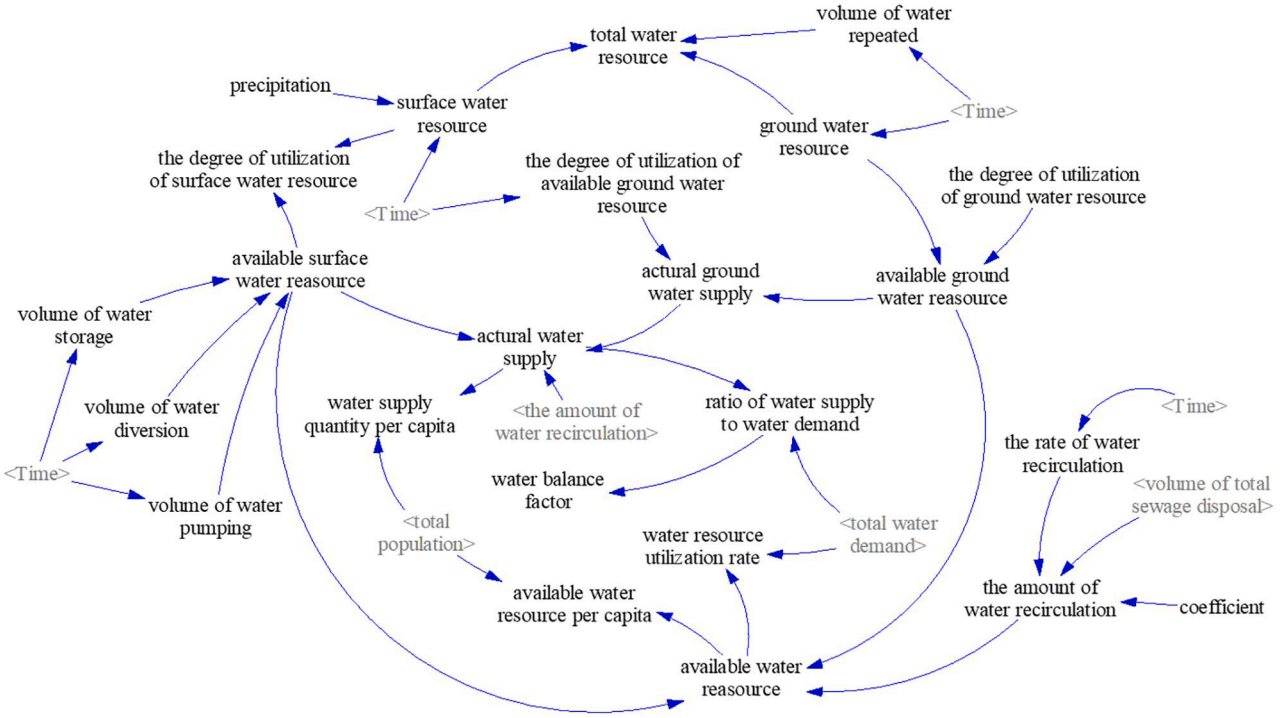
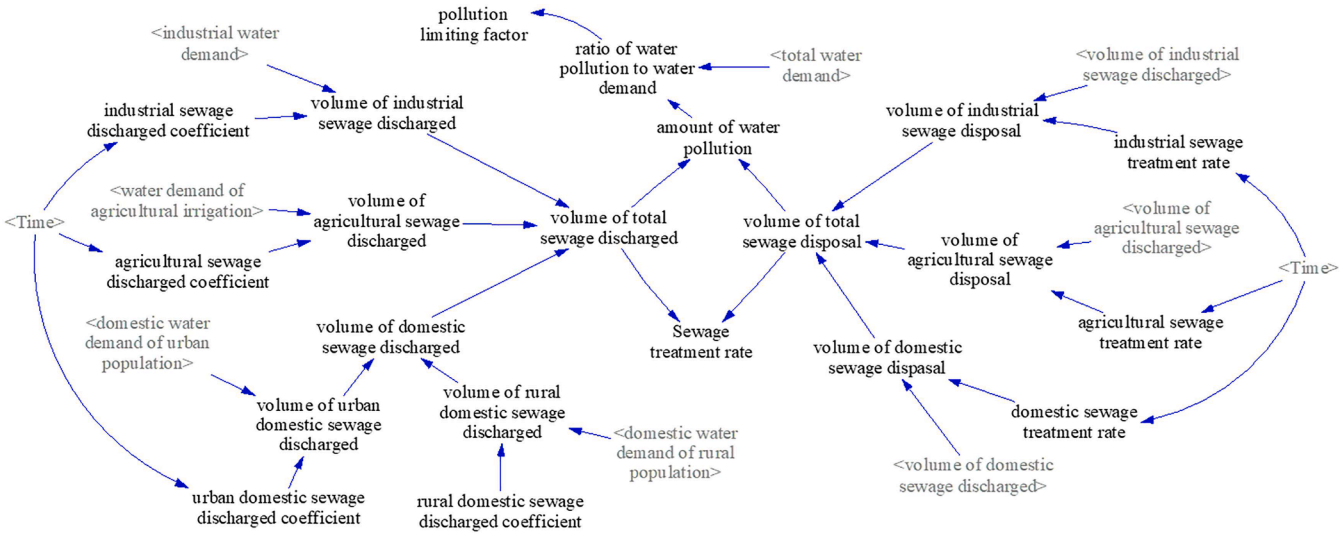


Fig. 6. System dynamics flowchart of water supply subsystem.



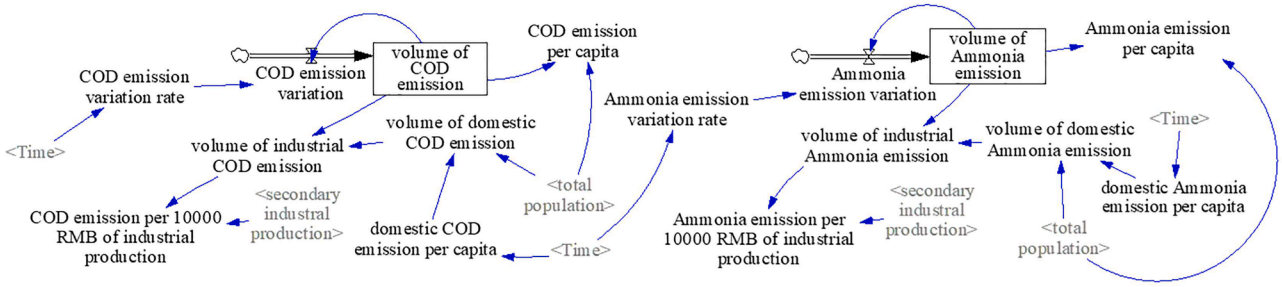


Fig. 7. System dynamics flowchart of water environment subsystem.

Table 3
Main variables and expressions in SD model.

Name	Unit	Equations
GDP	One hundred million yuan	=primary industrial production + secondary industrial production + tertiary industrial production
The proportion of secondary industry economy	%	= secondary industral production/GDP*100
Water Consumption Per 10,000 RMB of GDP	m^3/10,000 yuan	=Total Water Demand/GDP*10000
The rate of primary industrial production growth	dmnl	=WITHLOOKUP (Time, [(2005,-7)-(2025,24)],(2005,0.75),(2006,23.46),(2007,8.9),(2008,2.8),(2009,12.86),(2010,14.8),(2011,9.31),(2012,4.7),(2013,0.09),(2014,3.31),(2015,-5.77),(2016,-2.6),(2017,-4.35),(2018,2.86),(2019,8) (2025,8))
Rural population	people	=total population-urban population
Total Population	people	$=7314959+\int_0^t(\sum \text{BirthPopulationGrowth}-\sum \text{deathPopulationGrowth}+\sum \text{NetMigrationGrowth})dt$
Total water demand	m^3	= domestic water demand + FAHF water demand + industrial water demand + urban ecology water demand + urban public water demand + water demand of agricultural irrigation
Birth population growth	people	=total population*birth rate/1000*water balance factor
Green area	hectare	$=6118+\int_0^t(\sum \text{greenareagrowth})dt$
Available surface water reasource	m^3	= volume of water diversion + volume of water pumping + volume of water storage
Actural water supply	m^3	= available surface water reasource + actural ground water supply + the amount of water recirculation
Pollution limiting factor	dmnl	= IF THEN ELSE(ratio of water pollution to water demand>=0.04, 1.02 , 0.98)
Amount of water pollution	m^3	= volume of total sewage discharged- volume of total sewage disposal
Volume of total sewage discharged	m^3	= volume of agricultural sewage discharged + volume of domestic sewage discharged + volume of industrial sewage discharged
Sewage treatment rate	%	= volume of total sewage disposal/volume of total sewage discharged*100
Industrial sewage discharged coefficient	%	=WITHLOOKUP (Time, [(2005,6)-(2025,24)],(2005,17.68),(2006,18.46),(2007,16.3),(2008,23.35),(2009,21.78),(2010,23.74),(2011,16.63),(2012,16.8),(2013,14.54),(2014,13.19),(2015,9.08),(2016,8.04),(2017,7.97),(2018,5) (2022,5),(2025,5))

*Note: "With lookup" in Vensim software is a table function, which can be used to characterize non-linear relationship variables. "IF THEN ELSE" represents a function with selective variables. The state variable usually reflects the integration of the variable over time.

resources in Changchun in a timely manner, reflecting population and economic development to a certain extent. Total water supply is obtained by subtracting the repeated amount between them from the sum of surface water resources and groundwater resources. The amount of water supply directly affects the amount of water available, thereby impacting the regional WRCC.

4.1.5. Water environment subsystem

Sustainable development has gradually become the theme of world development. There is a need to tend to the quality of the water environment in order to improve theWRCC and maintain the sustainable development of regional water resource systems. Establishing a process relationship between the volume of sewage discharged and the volume of sewage disposed reflects the local pollution experienced in waterways. As the study area is an old industrial base, two water quality

Table 4
Results of model error validation.

Time	Total Population (10,000 Capita)			GDP (100 million)			Total Water demand (10^8 m^3)		
	Historical Data	Simulated Data	Error (%)	Historical Data	Simulated Data	Error (%)	Historical Data	Simulated Data	Error (%)
2005	731.50	737.461	0.815	1508.6	1508.9	0.020	16.73	16.76	0.168
2006	739.26	741.41	0.291	1741.2	1740.9	−0.017	18.95	18.96	0.047
2007	745.95	748.261	0.310	2089.0	2087.9	−0.053	23.05	23.06	0.065
2008	752.53	754.649	0.282	2561.9	2558.9	−0.117	19.66	19.66	0.011
2009	756.51	761.21	0.622	2848.6	2843.9	−0.165	20.92	20.94	0.106
2010	758.89	764.962	0.800	3329.0	3321.9	−0.213	18.69	18.70	0.076
2011	761.77	767.315	0.728	4003.0	3993.9	−0.227	23.71	23.73	0.084
2012	756.90	769.879	1.714	4456.6	4444.9	−0.263	22.51	22.56	0.203
2013	752.67	764.703	1.599	5003.2	4987.9	−0.306	24.03	24.07	0.160
2014	754.55	768.026	1.786	5342.4	5324.9	−0.328	23.55	23.60	0.221
2015	753.83	770.588	2.223	5530.0	5510.9	−0.345	24.00	24.07	0.283
2016	753.43	771.913	2.453	5928.5	5907.9	−0.347	24.45	24.57	0.498
2017	748.92	771.353	2.995	6530.0	6506.9	−0.354	24.39	24.54	0.618
Time	**Available Water Resources (10^8 m^3)**			**Amount of Water Pollution (10^4 ton)**			**Volume of COD Emission (ton)**		
	Historical Data	**Simulated Data**	**Error (%)**	**Historical Data**	**Simulated Data**	**Error (%)**	**Historical Data**	**Simulated Data**	**Error (%)**
2005	24.95	24.95	0.009	24088.2	23917.1	−0.710	81,542	81,543	0.001
2006	22.90	22.90	0.000	15546.1	15535.2	−0.070	83,237	83,238	0.001
2007	23.53	23.53	0.007	21177.2	21119.9	−0.270	77,342	77,344	0.003
2008	23.50	23.50	−0.002	17020.7	17115.4	0.557	66,633	66,635	0.003
2009	20.86	20.86	0.004	15792.8	15977.7	1.171	61,226	61,228	0.003
2010	25.93	25.93	0.000	12867.5	12987.4	0.932	64,011	64,013	0.003
2011	23.44	23.44	0.014	9821.8	9767.3	−0.555	43,711	43,713	0.005
2012	28.46	28.46	0.012	13493.0	13454.4	−0.286	42,848	42,850	0.005
2013	30.66	30.66	0.007	13954.8	14079.6	0.895	43,676	43,677	0.003
2014	23.17	23.18	0.029	12947.2	12942.0	−0.040	43,627	43,629	0.004
2015	25.07	25.07	0.012	11671.0	11694.7	0.203	42,142	42,144	0.005
2016	30.69	30.69	0.015	15424.5	15405.3	−0.124	25,536	25,538	0.009
2017	28.22	28.23	0.028	14446.3	14587.1	0.975	27,449	27,451	0.009

indicators were used: the chemical oxygen demand (COD) and ammonia nitrogen emissions.

4.2. Model verification and sensitivity analysis

Model error verification is the most direct way to verify whether a model meets the simulation effect. As the model uses a large number of variable parameters, the sensitivity of each parameter in the model is different. Therefore, to create a more representative error verification, six parameters with high sensitivity to the model output results were selected for calibration. The Monte Carlo method was used for sensitivity analysis, with each parameter randomly and uniformly sampled 1000 times within a set range. The model calibration parameters were selected between the 2005–2015 historical statistics, and the model verification parameters were selected between the 2016–2017 historical statistics. The simulation prediction time was from 2018 to 2025. A comparison of the simulated values with the historical values from 2005 to 2017 rendered a relative error of 0.19% during the calibration period and a relative error of 0.565% during the verification period (detailed results are shown in Table 4). The historical data selected for this model was available in greater detail, with a longer calibration period and a higher effectiveness. The selected simulation prediction time was short, and its error value was far less than the standard error value by 10%. This demonstrates that the model can achieve the simulation effect and has the ability to reflect the real system.

4.3. Evaluation of regional WRCC

This study simulates the change in the WRCC evaluation index over time in Changchun using an established SD dynamic feedback adjustment system. As each evaluation index has different magnitudes and standards, an improved fuzzy evaluation method was used to quantify the evaluation index. Based on the WRCC state classification method (Yang et al., 2010, 2019), the calculated evaluation value, λ, was divided into five categories to enable the quantitative and qualitative analysis of Changchun's WRCC. The WRCC from "weak" to "excellent" were: "weak" (λ between 0 and 0.2), "poor" (λ between 0.2 and 0.4), "normal" (λ between 0.4 and 0.6), "positive" (λ between 0.6 and 0.8), and "excellent" (λ between 0.8 and 1).

4.4. Scenario simulation

There are several different scenarios to promote the coordinated development of water resources, economy, and the environment in Changchun, and improve its WRCC. These scenarios are:

1) The status-quo scenario, maintaining the status quo social development model, and only considering the self-produced water resources in this area. This scenario was used as a reference scenario for others.
2) Scenario 1, strengthen the development of the secondary industry and fully develop the real economy. This is based on using Changchun's "13th five year" urban development requirements. The parameters relating to economic development (e.g., the growth rate of the secondary industry and the gross domestic product growth rate), were appropriately amplified and the feedback effects of the economic subsystem were obtained. Industrial water conservation was implemented concurrently to avoid the waste of water resources from the rapid development of the secondary industry. This is mainly achieved by increasing the rate of water recirculation and improving science and technology during the production process.
3) Scenario 2, promote the steady development of the primary industry, and is based on Scenario 1. In this scenario, the growth of primary industry production is appropriately increased to enable the full advantages of commodity grain bases. Whilst Changchun has a large cultivated area, it uses a single irrigation method dominated by flood irrigation, which reduces the utilization rate of water resources. In Scenario 2, agricultural water demand is reduced by changing irrigation methods and adopting drip irrigation technology.
4) Scenario 3, vigorously develop the tertiary industry and drive the market economy forward steadily, on the basis of Scenario 2. Compared with other industries, the tertiary industry in Changchun has great advantages with regard to its output value and its industrial structure. Changchun 13th Five Year Development Plan clearly encourages the development of the tertiary industry. To simulate this case, a 8% growth rate increase was applied to the tertiary industry. The water supply quantity per capita in Changchun is currently only 290 m^3/person, far lower than the average water supply of 436 m^3/person in the eastern cities of China. To avoid the tight supply and demand of water resources as a result of excessive rapid economic growth, Scenario 3 alleviates this issue by slightly adjusting population growth and urbanization rates.
5) Scenario 4, adjust the industrial structure, rationally allocate the economic proportion of primary, secondary and tertiary industries, and promote sustainable economic and social development. This scenario was developed based on Scenarios 1–3. For example, it considers the "Central City Group Water Diversion Project" to increase the supply of surface water resources. The construction of water-saving projects will also be strengthened concurrently to alleviate water shortage and improve the utilization rate of water resources.
6) Scenario 5, improve the quality of the water environment, based on Scenario 4. Continuously optimize the industrial structure and vigorously develop the environmental protection and energy saving industries. Promote the integration of agricultural machinery to accelerate grain production. Rapid economic development may also increase the risk of harm to the ecological environment. Water quality issues are now gaining increasing attention, and pollution control has become an important measure to improve the WRCC. This scenario is aimed at the current situation in Changchun with a focus on the treatment of urban black odor water. This is done through establishing a greater number of sewage treatment facilities and standardizing the sewage treatment process to increase the sewage treatment rate.

5. Results and discussion

5.1. Simulation results under the different scenarios

This study used Vensim software to establish a SD model to quantitatively simulate the evaluation index of WRCC in Changchun from 2018 to 2025, and analyze it's WRCC using IFCE qualitatively. The trends of WRCC in Changchun are discussed in this section under the six scenarios (Fig. 8).

The status-quo scenario maintained the status quo social development model, and there was a downward trend in the regional WRCC each year. Fig. 8, shows that the WRCC in the base year 2017 was in the normal carrying range and was projecting toward the poor carrying range by 2025. The current development model restricts the balance of water supply and demand, and cannot support large-scale economic and social development in the future. In addition, the study area was maintained at a normal carrying level by 2025. Although the values of various indicators were slightly lower, the basic development of Changchun in terms of economics, society, water resources and water environment is barely able to be maintained.

Scenario 1 allows for all characteristics of economic development of the old industrial base in Changchun. The proportion of the secondary industry's economy steadily increases from 48.9% to 55.5% between 2017 to 2025. Fig. 9a illustrates that the secondary industry's GDP reaches 680.09 billion yuan in 2025. This shows that Scenario 1 can effectively drive Changchun's economic growth. However, relying solely on industrial water savings does not address the increasing water demand pressure as a result of this growth. The ratio of water supply to

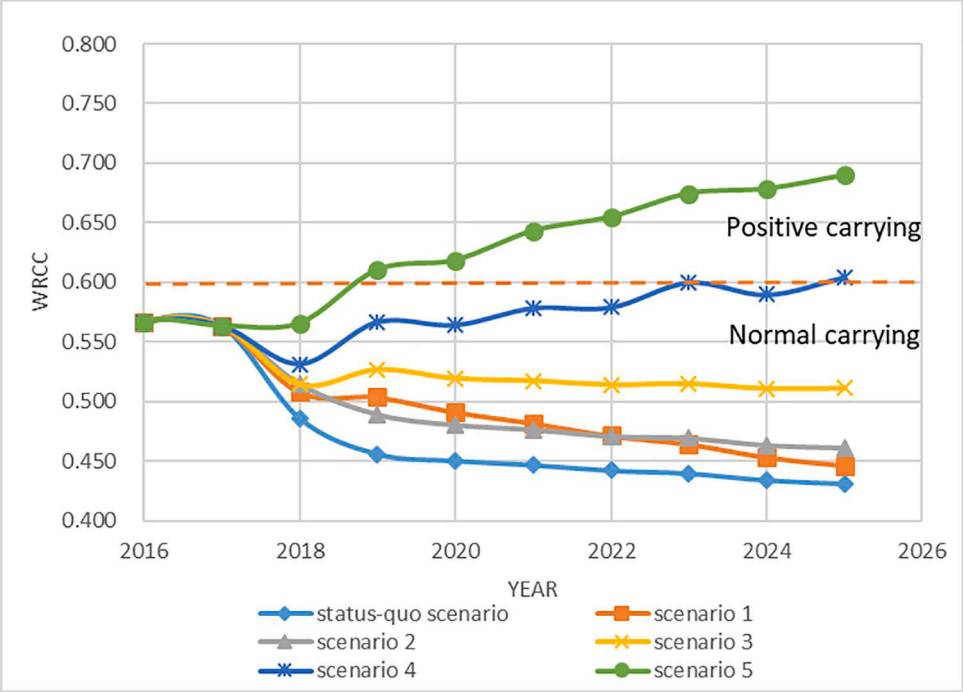


Fig. 8. The dynamic trends of the WRCC index in different scenarios.

water demand only reached 0.75 by 2025 in Fig. 9c, resulting in severe water shortages in Changchun. Fig. 8 shows that the WRCC is still within the normal carrying range, exhibiting a downward trend.

The primary industry plays a leading role in supporting rural livelihoods and economic growth (Guo et al., 2019). Changchun is located in the main agricultural area of northeast China plain, so it has adopted an agricultural support program to emphasize the economic development of Changchun. Scenario 2 is based on Scenario 1, focusing on the development of the primary industry and promoting the conservation of agricultural water. The economic proportion of the primary industry expands, and the growth rate of the primary industry increases by 7% each year. By 2025, the GDP of Changchun was expected to reach 12237 $\times$ 10^8. Based on the "Jilin Provincial Industry Water Quota Standard" and the "Changchun Municipal Government Development Plan", the irrigation water quota had reduced as a result of altered irrigation methods to promote water conservation in agricultural production. In the current model, the water quotas for paddy and vegetable fields were reduced by 225 and 150 m^3/hm^2, respectively, with all reduction values meeting the above standards. The results show (Fig. 8) that WRCC in Scenario 2 continues to be in a downward trend, however, the slope gradually slows.

According to the "Thirteenth Five-Year" Industrial and Information Development Plan of Changchun City (Changchun City Bureau of Industry and Information Technology, 2018), in Scenario 3, where the tertiary industry is vigorously developed, creating a "dolphin economy" to promote the modernization of the city. A dolphin economy is where a group of modern service industry enterprises that are smart and spiritual like dolphins, such as logistics, online sales, etc. The tertiary industry's average economic proportion was 51%, and by 2025, the regional GDP reaches 12.29 billion yuan. By 2025, the total population decreases by 5.4% compared to the status quo scenario, and the urban population increases by 5.5% (Fig. 9b). The results (Fig. 8) show that the WRCC index steadily recovered from a continuous decrease and ascended to a stable carrying status.

Scenario 4 which combines the above scenarios, focuses on adjusting the industrial structure and promoting the development of the secondary and tertiary industries. Changchun is a water scarce area, where its water resources per capita are only a quarter of China (Dai et al., 2012). However, the diversion of the surrounding surface water resources can effectively address this water shortage issue (Zhu et al., 2014). The plan takes into account the "Central City Group Water Diversion Project", which increases surface water supply by 0.06 $\times$ 10^8 m^3/a. It will also continue to increase the construction of water-saving facilities and improve the utilization of water resources. Whilst saving water for domestic water use and production, it also advocates for ecological water saving, where the quota of green area irrigation is reduced by 500 m^3/hm^2. Through these mechanisms, the water supply to demand ratio increases significantly (Fig. 9c). Scenario 4 shows that the WRCC greatly improves and increases each year. By 2022, the carrying level improves from normal to positive level.

Scenario 5, focuses on improving the quality of the water environment. The increase in water demand for energy generation may conflict with the objectives of river conservation according to the Water Framework Directive (Kuriqi et al., 2017). Thus, effective remediation strategies for reducing the emissions and background levels of pollution in the environment are urgently needed (Chen et al., 2017) In Scenario 5, there is a 3% and 6% reduction in the industrial and agricultural sewage discharge coefficients, respectively, to control sewage discharge from the source. The sewage treatment rate increases by 6%, whilst sewage volume has reduced by approximately 1.03 $\times$ 10^8 t. In this scenario, the water environment system develops healthily and steadily (Fig. 9d). Fig. 7 shows that since 2019, Changchun has been positive carrying, with the carrying capacity steadily improving.

5.2. Comparison of scenarios

The status quo scenario demonstrates a continual increase in the conflict between the five subsystems, with a waning development tendency. As such, continuing in the current development scenario cannot meet the sustainable development needs of Changchun.

In Scenario 1, economic growth was significantly higher than the status quo scenario, specifically, and by 2025, the city's GDP increased by 6.7%; more than the status quo scenario. However, water supply and demand capacity remained low, which does not confirm the potential for sustainable development. During the simulation period, although the WRCC of Scenario 1 had slightly improved compared with the status quo

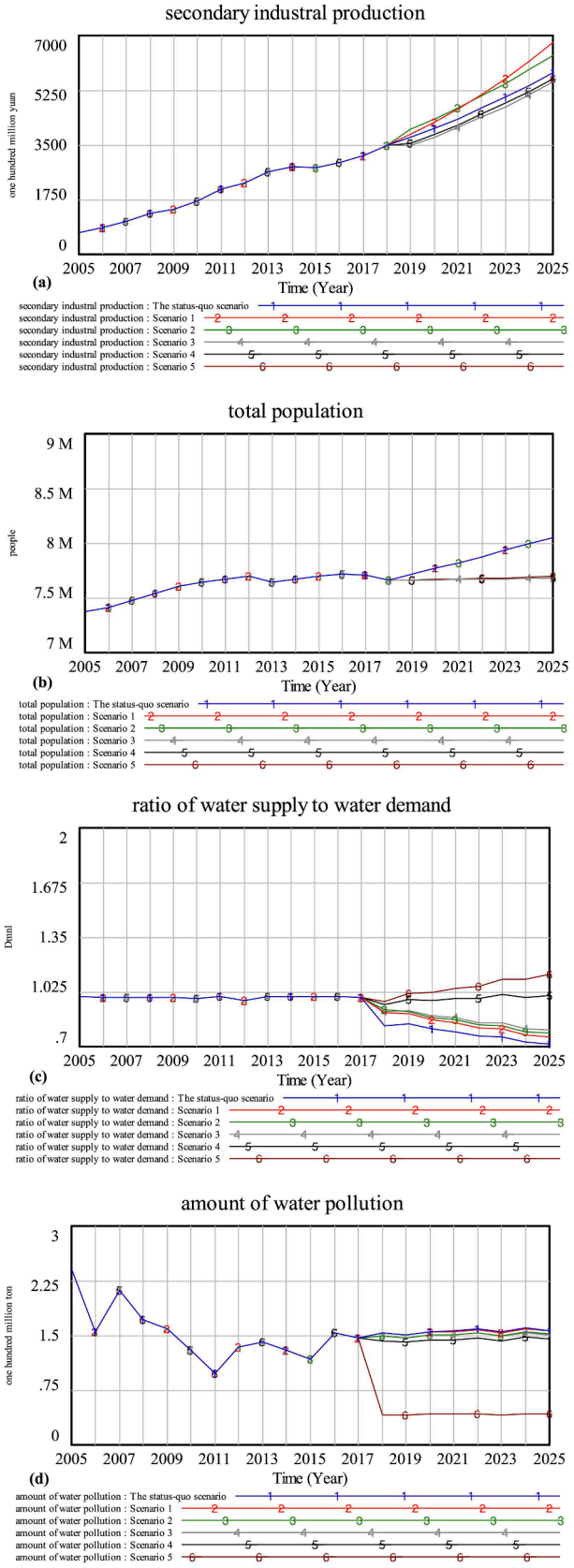


Fig. 9. The dynamic trends of the representative indicators in different scenarios. (a) secondary industrial production (b) total population (c) ratio of water supply to water demand (d) amount of water pollution.

scenario, there was still an overall decreasing trend. This demonstrates that the industrial water savings measures are able to alleviate the pressure from a large increase in water demand as a result of excessively rapid industrial growth; however, this was not an obvious improvement. Simply increasing industrial development capacity is insufficient to maintain the economic development of Changchun.

Despite a significant improvement in WRCC in Scenario 2 compared to the status quo scenario, the overall level of improvement is not significant compared with Scenario 1 (Fig. 8). Research has shown that the tertiary industry product, gross domestic product, and secondary industry product were driving factors of urban land sprawl in Changchun from 1990 to 2015 (Chen et al., 2018). Therefore, the reason for the insignificant improvement in scenario 3 may be it ignores the driving advantages of the tertiary industry and creates a conflict between increasing supply and demand.

Based on the results of Scenario 3, the WRCC and the ratio of supply to demand of water resources have improved significantly compared with the status quo scenario. This indicates that tertiary industry development can effectively improve the WRCC in Changchun, and controlled population growth may alleviate the tension between water supply and demand. A comparison of the results from Scenario 1 which focuses on industrial development and Scenario 2 which focuses on accelerating agricultural development, demonstrates that Changchun is more suited to the development model with the secondary and tertiary industries as the mainstay, and the primary industry as the supplement. Remarkably, the ratio of water supply to demand is still unable to meet the needs of sustainable development. A rational industrial structure conducive to the sustainable utilization of water resources is required to facilitate sustainable development (Bao et al., 2006; Zhang and Guo, 2010).

An adjustment of the industrial structure, generated a better economic performance in Scenario 4 than in other scenarios, however, it was slightly worse than Scenario 1. Although Scenario 4 was the most effective scenario in terms of improving the WRCC, the water environment system continues to deal with huge challenges. The limiting effect of water quality on water resources systems has gradually gained attention (Fu et al., 2017). Even if the quantity of water resources in a region is abundant, once the water quality is classified as Class IV or Class V, the WRCC is also weak. Specifically, the ratio of water pollution to water demand was 1.07 times higher than that in the status quo scenario. This illustrates that whilst re-balancing water supply and demand balance may improve the WRCC, but a greater extent of change is needed. The WRCC is both affected by the shortage of water resources, and the quality of the water environment.

The consideration of multiple perspectives in Scenario 5 produces a marked improvement in the WRCC. In this scenario, within the context of serious agricultural pollution in rural areas, sewage treatment facilities were built to mitigate the intensifying water pollution. As a result of these facilities, the amount of water pollution decreases from 1.56×10^8 t in the status quo scenario to 0.42×10^8 t in Scenario 5. The agricultural wastewater treatment rate increased from 40% to 70% compared to the other scenarios. The evaluation of WRCC in Changchun was at a positive carrying level in Scenario 5. It's suggested that the WRCC in Changchun should be improved by saving water, adjusting industrial structure, improving the pollution control, and increasing the amount of recycled water (Wang et al., 2017). This shows that the regional WRCC was a result of the combined effects of economics, society, water resources and water environment. A key part of the success of Scenario 5 is addressing protection of the water environment whilst promoting production. Overall, Scenario 5 maximizes steady economic and social development in Changchun.

5.3. Policy implications

As an old industrial base in Northeast China, Changchun has unique economic advantages for its industrial economic development.

However, excessive rapid industrial production is likely to result in water shortages due to burgeoning water demand. The 13th Five Year Plan of Changchun highlights that it is necessary to promote the economic and industrial structure adjustment of Changchun, and use the "dolphin economy" to promote the economic development of the tertiary industry. The simulation of six different scenarios confirms that vigorously promoting the secondary and tertiary industries, and maintaining a steady primary industry can maximize the "the positive and rapid" development of Changchun. However, economic growth will inevitably increase the demand for water resources. Although the primary industry occupies a relatively small proportion of the economy in Changchun, agricultural water consumption accounts for more than half the total water consumption. Under such circumstances, appropriately reducing the industry's water quota and increasing the rate of water resources utilization are important measures to effectively reduce total water demand. The simulation results show that although water demand from agriculture is trending downward, it continues to account for a large proportion of the total water demand. The reduction of irrigation water cannot fundamentally eliminate the agricultural water shortage. At present, agricultural development is mainly dependent on surface irrigation and traditional sprinkler irrigation in areas throughout northern China (Li et al., 2012). The antiquated agricultural irrigation infrastructure and inefficient irrigation methods are the primary factors underpinning the high water consumption by the agricultural industry in Changchun (Zhang et al., 2014). As such, local government should accelerate the construction of water-saving facilities and substitute outdated irrigation methods. Promote water-saving technologies (e.g., drip irrigation and sprinkler irrigation) and strengthening engineering measures (e.g., channel anti-seepage) will improve the effective irrigation coefficient.

In recent years, in response to the Changchun's water shortage, the Changchun Municipal Government has strongly supported the construction of water diversion projects (Chen et al., 2019). They plan to address the shortage of surface water supply by drawing on water from the Songhua River. This form of water supply will eventually become the main urban water supply in the future. To alleviate the urban water supply shortage, urban water management departments continue to increase the construction of water reuse facilities. Based on the results, the capacity of Changchun to treat circulating water is relatively weak. The poor planning associated with the circulating water infrastructure has greatly restricted the use of recycled water. At present, the rate of water recirculation in Changchun is only 12%, far lower than the national average. As such, effectively using water diversion projects and strengthening the construction of reclaimed water pipeline networks may be effective solutions to bridge the gap between water supply and demand in Changchun.

In recent years, there has been greater attention given to the protection of the ecological environment, with an understanding that there is serious agricultural and industrial pollution in the area (Ma et al., 2017). Due to the unrestricted use of pesticides and fertilizers, the ammonia content of surface water in most areas exceed the standard. In addition, the rural production and domestic water treatment facilities are outdated, with a 40% sewage treatment rate. This violates the concept of green ecology and seriously hinders the sustainable development of the regional economy. Improving the urban–rural integration of sewage treatment facilities and strengthening the rural ecological environment may greatly promote the active development of urban environment. Upgrading urban–rural integrated sewage treatment facilities and strengthening rural ecological environment governance may greatly enhance the active development of urban environment.

5.4. Limitations and future research directions

In general, this study chooses an improved fuzzy comprehensive evaluation method to solve the fuzziness and uncertainty of influencing factors. The addition of the AHP method can effectively reduce the interference of human factors and make the weight selection more reasonable. At the same time, the improved fuzzy comprehensive evaluation method is combined with the SD dynamic simulation model to achieve qualitative and quantitative evaluation.

However, there are some limitations exist in this research.

(1) Although this study involves as many relevant influencing factors as possible, it still cannot cover all influencing factors. Some influencing factors have been simplified or not taken into account, for example, water price changes and the impact of climate change on the WRCC. (Yang et al, 2019; Cai et al., 2011)
(2) Due to the confidentiality of some data (such as wastewater treatment and discharge data), it is impossible to collect more detailed information, which affects the accuracy of the relevant model simulation. Therefore, in order to ensure the accuracy of the simulation results, the simulation time is set at 8 years. If more detailed data can be collected in the future, I hope to supplement and improve on this basis.
(3) The SD model and the improved fuzzy comprehensive evaluation method used in this study are carried out in stages. In the later research, the improved fuzzy comprehensive evaluation method can be integrated into the dynamic system of the SD model to achieve true coupling. We believe that the coupled model is more capable of dynamic evaluation.

In order to achieve a more comprehensive research on the WRCC, cross-domain methods and the coupling of SD models and numerical models can be used as the focus of future research.

6. Conclusions

This study used a SD model to establish five mutually restricted subsystems, confirming the reliability of the model using the model's error correction and sensitivity analysis. The evaluation selected 17 influential factors with high sensitivity, and the IFCE method was selected for dynamic evaluation of the model. The SD simulation results directly and quantitatively illustrated the change to WRCC based on various influencing factors, and provided a qualitative evaluation in combination with IFCE. These methods were used in conjunction with six formulated scenarios to simulate the dynamic trend of WRCC under different social development modes. The results of each scenario were compared and discussed to select the most suitable development model for Changchun and formulate a conclusion.

The study proposes six scenarios for the reference of decision makers. The simulations show that if there are no changes to the existing development model, the WRCC of Changchun will decrease each year over the next five years, maintaining a lower normal carrying range. Scenarios 1–3 show that Changchun is better suited to an economic development model with the secondary and tertiary industries as the mainstay, and a supplementary primary industry. The WRCC can significantly improve by vigorously developing energy conservation and environmental protections and promoting the construction of a conservation-oriented society. Scenario 4 showed that the balance between water supply and demand is an important factor affecting the WRCC in Changchun. Focusing on the reality of water shortage, water diversion projects may alleviate the pressure of urban construction. Scenario 5 clearly demonstrates that protection of the water environment is also an important factor affecting the WRCC. Scenario 5 ensures the harmonious development of economy, society, water resources and water environment, which is the suitable development scenario. The results also illustrate that outdated irrigation facilities and inefficient irrigation methods are the main drivers generating huge agricultural irrigation water consumption. Therefore, under the interaction mechanism of economy-population-water resources-water environment, Changchun urgently needs to adjust the industrial layout, solve the problem of water supply and demand balance and water environment

pollution. New water-saving methods such as drip irrigation and micro drip irrigation should be promoted. Strengthening the construction of rural sewage treatment facilities in Changchun is also urgently needed. There should be emphasis from the government on the integrated construction of urban and rural sewage networks to ensure the sustainable development of the water environment.

The research exemplifies the superiority of the integrated method combining the IFCE and the SD methods. This integrated method may overcome intrinsic limitations with regard to quantitative indicators of sustainability and create a more credible analysis of future development trends. The research direction for the future can focus on the coupling of dynamic models and mathematical models, rather than a staged combination. Establish a coupling model based on the SD model and the mathematical model to realize the real dynamic feedback assessment between the water resources subsystems.

CRediT authorship contribution statement

Ge Wang: Conceptualization, Methodology, Formal analysis, Writing - original draft. **Changlai Xiao:** Data curation, Funding acquisition, Project administration. **Zhiwei Qi:** Writing - review & editing, Formal analysis. **Fanao Meng:** Investigation. **Xiujuan Liang:** Data curation, Funding acquisition, Project administration, Supervision.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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